

Deep Learning: Homework 1

1. Your task is to find a linear approximation of the function $\sqrt{1+x}$, $x \in [0, 1]$. Your homework should contain the following steps:

- a) Generate $N = 10000$ random numbers from $[0, 1]$:

$$x_1, x_2, \dots, x_N \in [0, 1],$$

and then obtain their labels: $y_i = \sqrt{1+x_i}$, $i = 1, 2, \dots, N$.

- b) Do linear regression on your generated data using the closed form solution.
- c) Do linear regression on your generated data using the library *sklearn*.
- d) Do linear regression on your generated data implementing the gradient descent algorithm by yourself.
- e*) Do linear regression on your generated data using tensorflow.
- f) Sketch the graphs of all approximations on one graph.
- g) Compare all solutions with the first degree Taylor approximation of the function $\sqrt{1+x}$.
- 2*. a) How will you define polynomial regression inspired from linear regression?
- b) Can you implement the polynomial regression using linear regression?